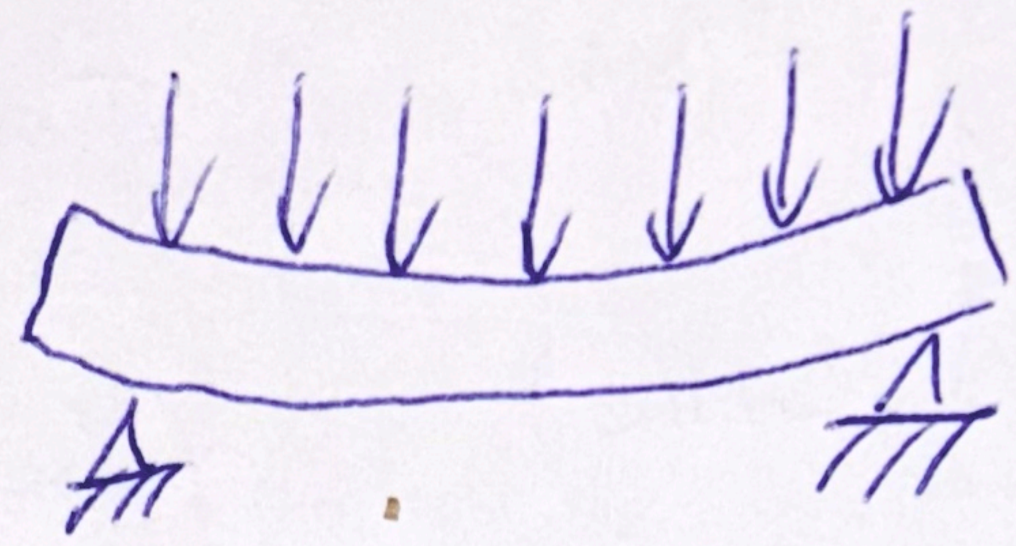
Bending beam



- Forces in the beam

  Consider a cross-section

  Convention : forces act left → right

  Newton III : reaction force righ → left opposite

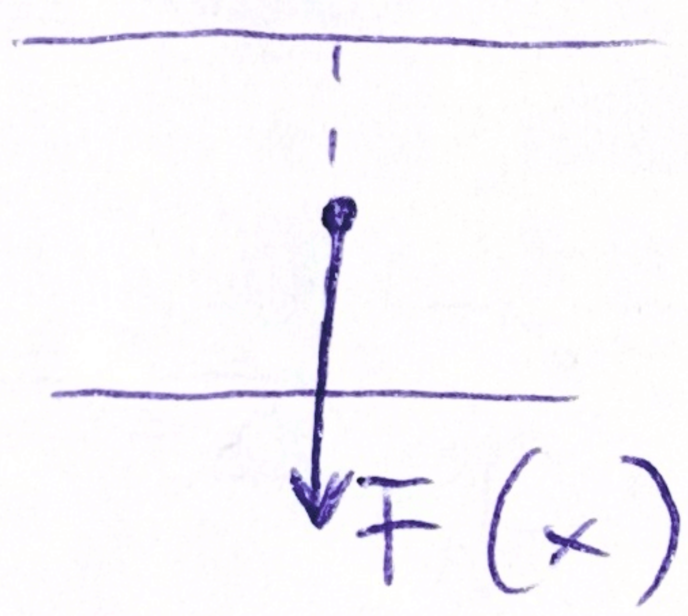
- Horizontal forces

  Net horizontal force is 0

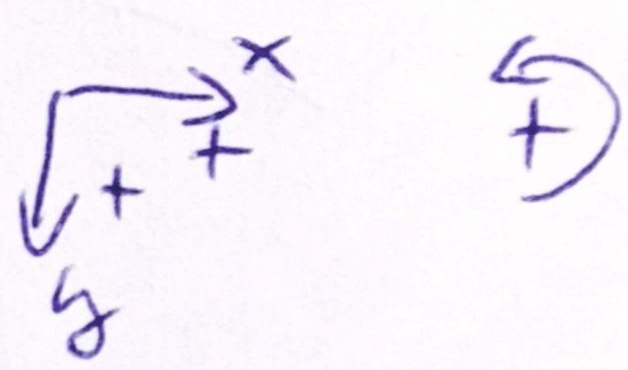
  (if horizontal beam, vertical load)

  Characterize by applied torque (to the right) $M(x)$

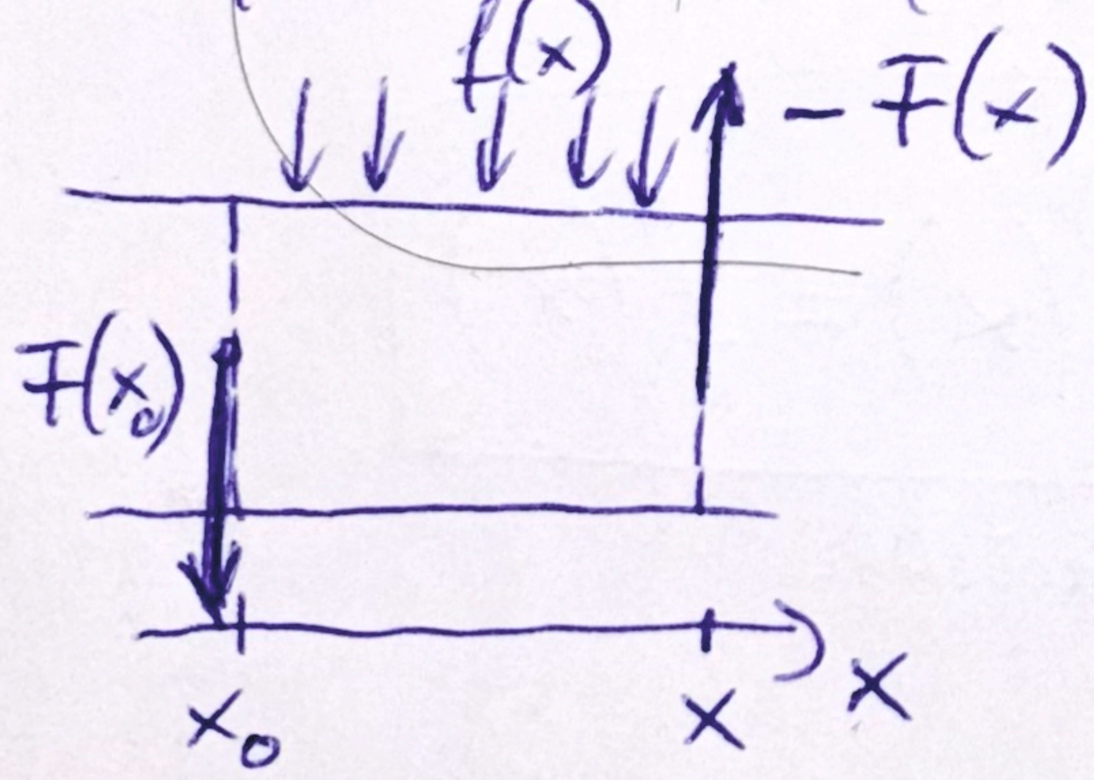
- Vertical (shear forces) + Load $f(x)$



- Sign conventions



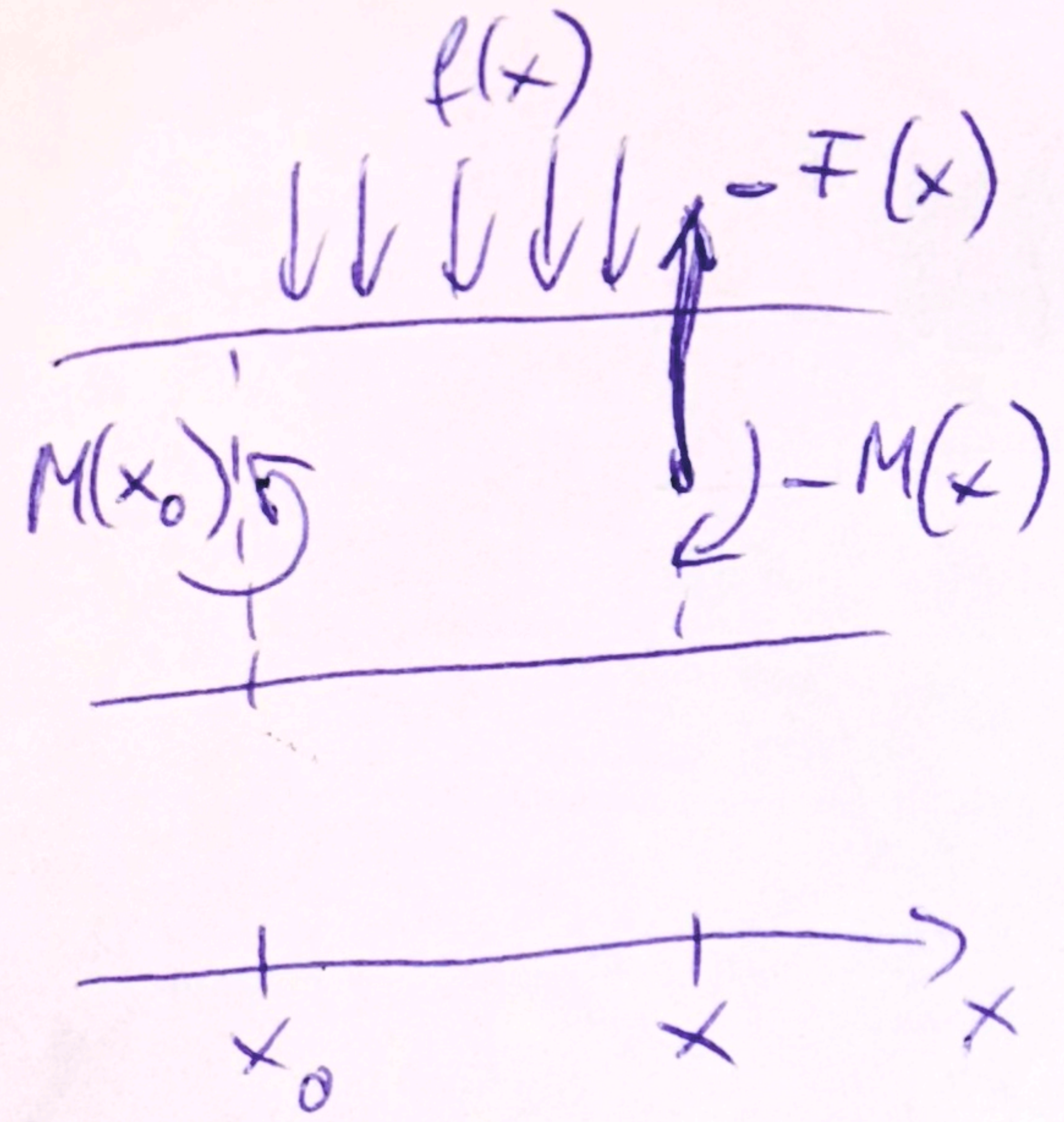
- Force and torque equilibrium for a section of a beam (assuming no bending, rigid limit)



$$F(x_0) - F(x) + \underbrace{\int_{x_0}^{x} f(x')\,dx'}_{\text{total load}} = 0$$

$$\frac{d}{dx} \downarrow$$

$$+\ \boxed{\frac{dF}{dx} = f(x)}$$



Torque wrt. x_0

$$M(x_0) - M(x) + (x - x_0)F(x) - \int_{x_0}^{x} (x' - x_0) f(x')\,dx' = 0$$

$$M(x) = M(x_0) + (x - x_0)F(x) - \int_{x_0}^{x} (x' - x_0) f(x')\,dx'$$

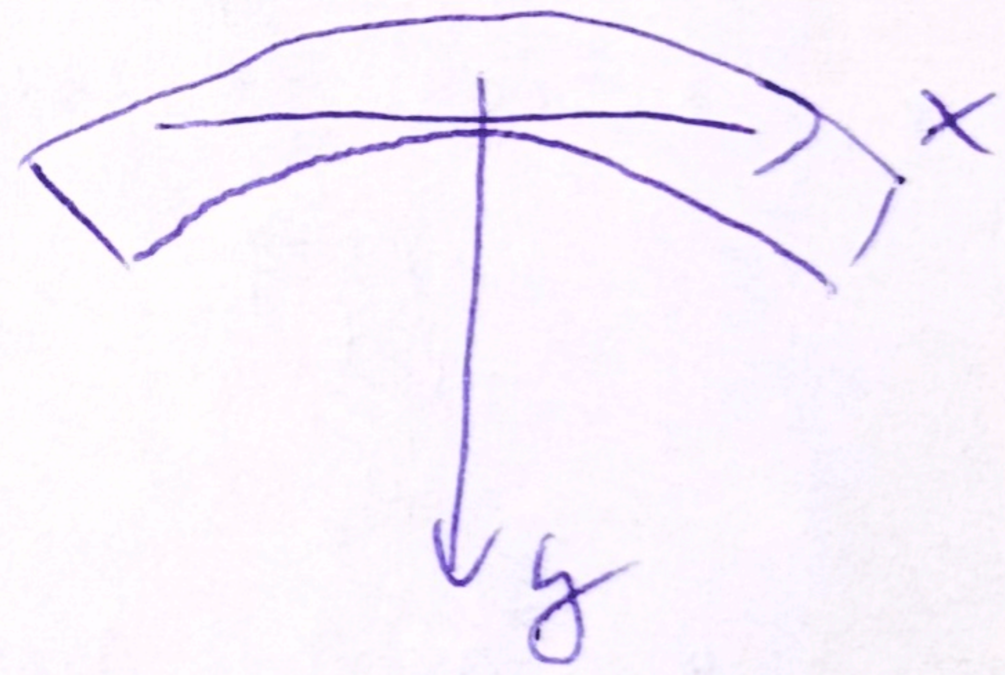
$$\Downarrow \frac{d}{dx}$$

$$\frac{dM}{dx} = F(x) + (x - x_0)\underbrace{\frac{dF}{dx}}_{f(x)} - (x - x_0) f(x) = F(x)$$

$$\Downarrow \frac{d}{dx}$$

$$\boxed{\frac{d^2M}{dx^2} = \frac{dF}{dx} = f(x)}$$

Curvature of the beam $\propto \frac{d^2y}{dx^2} \propto$ bending moment



$$E \cdot I \, \frac{d^2y}{dx^2} = M(x)$$

Young's modulus material property

2nd moment of cross-section shape property

$$\Longrightarrow \quad \boxed{E \cdot I \cdot y^{(4)}(x) = f(x)}$$

beam equation

Physical meaning

$y(x)$ position

$y'(x)$ slope

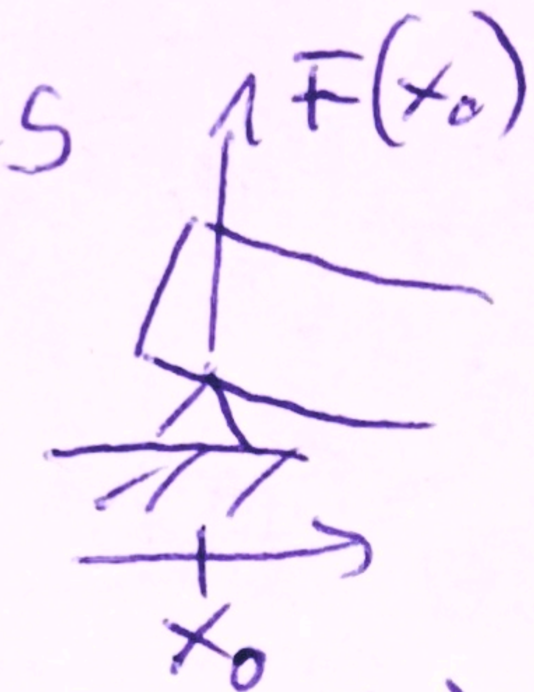
$y''(x)$ curvature $\propto$ bending moment $M(x)$

$y'''(x)$ $\propto$ shear force $F(x)$

Boundary conditions

- Simple support

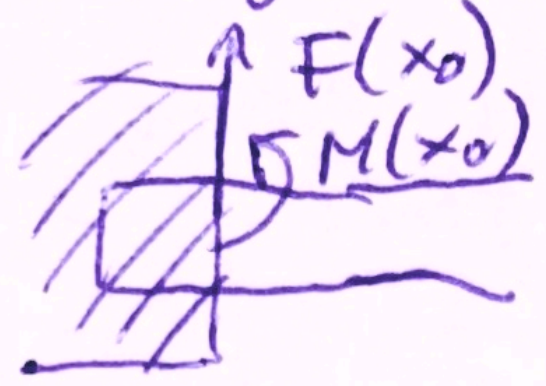
  rotates freely $\Rightarrow M(x_0)=0$



$$y(x_0)=0$$
$$y''(x_0)=0$$

- Clamped

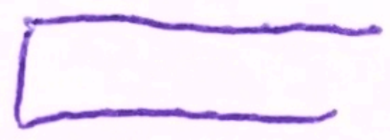
  no rotation



$$y(x_0)=0$$
$$y'(x_0)=0 \quad \text{(horizontal)}$$

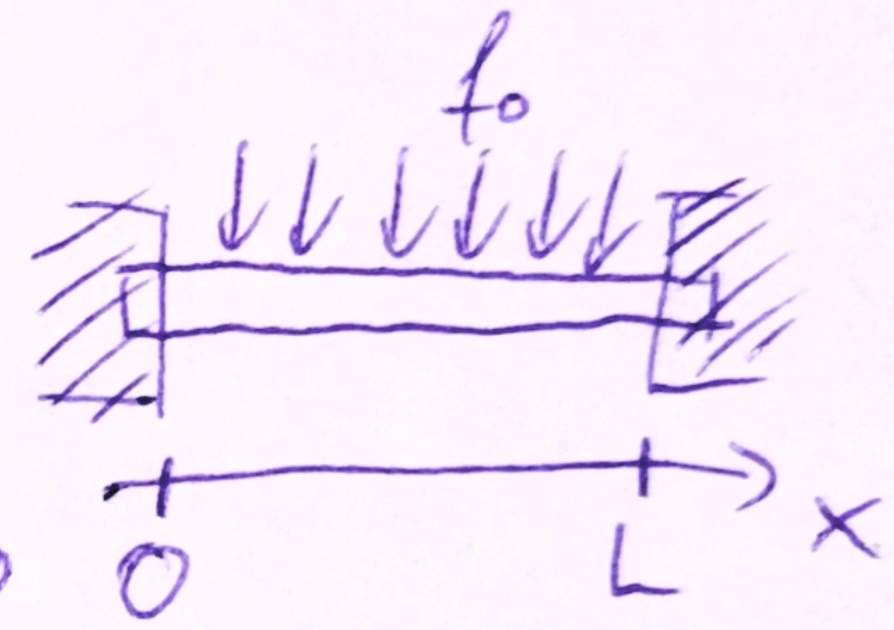
- Free

  no force $\quad F(x_0)=0 \quad M(x_0)=0$



$$y''(x_0)=0$$
$$y'''(x_0)=0$$

4.) Uniform load, two clamped ends



$$y^{(4)}(x)=\frac{f_0}{E\cdot I}=k \qquad y(0)=0 \quad y(L)=0 \qquad y'(0)=0 \quad y'(L)=0$$

Homogeneous solution

$y^4(x)=0 \qquad y(x)=e^{\lambda x} \Rightarrow \lambda^4=0 \Rightarrow \lambda=0$ 4 times root

$y_1(x)=e^{\lambda x}=1$, other solutions from multiplying with x

$y_2(x)=x$, $y_3(x)=x^2$, $y_4(x)=x^3$

Particular solution

$r(x)=k$ o'th order polynomial, try $y_p(x)=k_1$

same as $y_1(x)\rightarrow$ multiply by $x\rightarrow$ same as $y_2(x)\rightarrow$

$\rightarrow$ (repeat) $\dots\rightarrow y_p(x)=k_1 x^4$

$$y_p^{(4)}(x)=k_1\cdot 4\cdot 3\cdot 2=24k_1=k \Rightarrow y_p(x)=k\frac{x^4}{24}$$

General solution:

$$y(x)=c_1+c_2x+c_3x^2+c_4x^3+\frac{kx^4}{24}$$

Boundary conditions (IVP)

$$y(0) = c_1 = 0$$

$$y'(x) = c_2 + 2c_3 x + 3c_4 x^2 + k\frac{x^3}{6}$$

$$y'(0) = c_2 = 0$$

$$y'(L) = 2c_3 L + 3c_4 L^2 + k\frac{L^3}{6} = 0 \quad / :L$$

$$y(L) = c_3 L^2 + c_4 L^3 + k\frac{L^4}{24} = 0 \quad / :L^2$$

$$2c_3 + 3c_4 L + k\frac{L^2}{6} = 0 \qquad (1)$$

$$c_3 + c_4 L + k\frac{L^2}{24} = 0 \qquad (2)$$

$$(2) \Rightarrow c_3 = -\left(c_4 L + k\frac{L^2}{24}\right)$$

$$(1) \quad -2c_4 L - 2\frac{L^2}{24}k + 3c_4 L + \frac{L^2}{6}k = 0 \qquad /:L$$

$$c_4 = -k\frac{L}{12} \qquad c_3 = k\frac{L^2}{24}$$

$$y(x) = k\left[\frac{L^2}{24}x^2 - \frac{L}{12}x^3 + \frac{x^4}{24}\right]$$

2.7/12

$y'' + 4y = \underbrace{-12\sin(2x)}_{r(x)}$ $\qquad$ $y(0) = 1.8\ ;\ y'(0) = 5.0$

$y(x) = y_h(x) + y_p(x)$

1. Solve hom. eq.

$y'' + 4y = 0$ $\qquad$ const. coeffs. $\rightarrow$ $y = e^{\lambda x}$

char. eq. $\lambda^2 + 4 = 0$

$\lambda_{1,2} = \pm\sqrt{-4} = \pm 2i$

$y = \tilde{C}_1 e^{2ix} + \tilde{C}_2 e^{-2ix} = C_1 \cos(2x) + C_2 \sin(2x)$

2. Find the part. sol $y_p(x)$ using the Method of Undetermined Coeffs

$r(x) \rightarrow$ $A\cos(2x) + B\sin(2x)$?

Duplication! $\rightarrow$ multiply by x!

Ansatz: $y_p = Ax\cos(2x) + Bx\sin(2x)$

$y_p' = A\cos(2x) - 2Ax\sin(2x) + B\sin(2x) + 2Bx\cos(2x)$

$y_p'' = -2A\sin(2x) - 2A\sin(2x) - 4Ax\cos(2x) +$
$+ 2B\cos(2x) + 2B\cos(2x) - 4Bx\sin(2x) =$
$= -4A\sin(2x) + 4B\cos 2x \underbrace{- 4x(A\cos(2x) + B\sin(2x))}_{-4y_p}$

Substitute & simplify

$-4A\sin(2x) + 4B\cos 2x = -12\sin(2x)$

$\Rightarrow B = 0$

$-4A = -12 \rightarrow A = 3 \rightarrow y_p = 3x\cos(2x)$

2.7/18

$(D^2+2D+101)y = 17\sin x - 37\sin 3x$ $\quad y(0)=6.6$

$y'(0)=-2.2$

$y'' + 2y' + 10y = \underbrace{17\sin x}_{y_{p1}} \underbrace{-37\sin 3x}_{y_{p2}}$

1. Hom. eq

$\lambda^2 + 2\lambda + 10 = 0$

$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-2 \pm \sqrt{4-40}}{2}$ $\nearrow -1+3i$, $\searrow -1-3i$ $\quad (\lambda = \alpha \pm \beta i)$

$y_h = e^{\alpha x}(C_1\cos(\beta x) + C_2\sin(\beta x))$

$y_h = e^{-x}(C_1\cos(3x) + C_2\sin(3x))$

2. part. sol (MUC + superposition principle)

$y_{p_1} = A\cos + B\sin x$ no duplication (since no e^{-x} term)

substitute derivatives

$-A\cos(x) - B\sin x + 2[-A\sin x + B\cos x] +$

$+ 10[A\cos x + B\sin x] = 17\sin x$

$\cos x$: $9A + 2B = 0$

$\sin x$: $-2A + 9B = 17$

$\left.\right\}$ $A = -0.4$, $B = 1.8$

$\rightarrow y_{p1} = -0.4\cos x + 1.8\sin x$

$y_{p_2} = 6\cos(3x) - \sin(3x)$ (similarly

$y = e^{-x}(C_1\cos(3x) + C_2\sin(3x)) - 0.4\cos x + 1.8\sin x + 6\cos(3x) - \sin(3x)$

from initial conditions $C_1 = 1.0$ and $C_2 = 0$

$\Rightarrow y(x) = e^{-x}\cos(3x) - 0.4\cos x + 1.8\sin(x) + 6\cos(3x) - \sin(3x)$

$$y(x) = C_1 e^{-2x} + C_2 x e^{-2x} - \frac{1}{4} e^{-2x} \sin(2x)$$

3. Initial conditions

$$1 = y(0) = C_1 e^{0} + C_2 \cdot 0 \cdot e^{0} - \frac{1}{4} e^{0} \sin 0$$

$$1 = C_1 \cdot 1 + 0 - 0$$

$$C_1 = 1$$

$$y'(x) = -2 C_1 e^{-2x} + \left[C_2 e^{-2x} - 2 C_2 x e^{-2x}\right] - \frac{1}{4}\left[-2 e^{-2x} \sin 2x + e^{-2x}(2\cos(2x))\right]$$

$$-1{,}5 = y'(0) = -2 C_1 + C_2 - \frac{1}{4}[0+2]$$

$$-1{,}5 = -2 C_1 + C_2 - 0{,}5 \rightarrow \quad -1{,}5 = -2 + C_2 - 0{,}5 \rightarrow \quad C_2 = 1$$

$$y(x) = e^{-2x}\left(1 + x - \frac{1}{4} \sin(2x)\right)$$

2.7 /14

$y'' + 4y' + 4y = \underbrace{e^{-2x} \sin 2x}_{r(x)}$ $\qquad y(0) = 1,\ y'(0) = -1{,}5$

$y(x) = y_h(x) + y_p(x)$

1. Solve hom eq.

$y'' + 4y' + 4y = 0$

$\lambda^2 + 4\lambda + 4 = 0$ $\quad \lambda_1 = -2$

$(\lambda + 2)^2 = 0$ $\quad \lambda_2 = -2$

Rule for repeated roots: $y_p = C_1 e^{\lambda x} + C_2 x e^{\lambda x}$

$y_p = C_1 e^{-2x} + C_2 x e^{-2x}$

no duplication

no modification rule is needed!

2. Find $y_p(x)$ using MUC

$r(x) \to \quad y_p(x) = e^{-2x} (A \cos(2x) + B \sin(2x))$

$y_p'(x) = -2e^{-2x}(A \cos(2x) + B \sin(2x)) +$
$e^{-2x}(-2A \sin 2x + 2B \cos(2x))$

$= e^{-2x}[(-2A + 2B)\cos(2x) + (-2A - 2B)\sin(2x)]$

$y_p'' = e^{-2x}[-8B \cos(2x) + 8A \sin(2x)]$

Subs. into the diff. eq and divide by e^{-2x}

$-8B \cos(2x) + 8A \sin(2x) + 4[(-2A + 2B)\cos(2x) + (-2A - 2B \sin(2x))]$
$+ 4[A \cos(2x) + B \sin(2x)] = 1\sin(2x)\ (+ 0 \cos(2x))$

$\cos(2x)$ terms: $-8B + 4(-2A + 2B) + 4A = 0$

$-4A = 0 \to A = 0$

$\sin(2x)$ terms: $8A + 4(-2A - 2B) + 4B = 1$

$-4B = 1$

$B = -\frac{1}{4}$

$y_p(x) = -\frac{1}{4} e^{-2x} \sin(2x)$

$y(x) = C_1 \cos(2x) + C_2 \sin(2x) + 3x\cos(2x)$ (general sol.)

3. Initial conditions

$$y'(x) = -2C_1 \sin(2x) + 2C_2\cos(2x) + 3\cos(2x) - 6x\sin(2x)$$

$$5.0 = y'(0) = \underbrace{-2C_1 \sin 0}_{0} + 2C_2 \underbrace{\cos 0}_{1} + 3\underbrace{\cos 0}_{1} - \underbrace{6 \cdot 0 \cdot \sin 0}_{0}$$

$$5.0 = 2C_2 + 3 = 5$$

$$2C_2 = 2 \rightarrow C_2 = 1$$

$$1.8 = y(0) = C_1 \cdot 1 + C_2 \cdot 0 + 3 \cdot 0 = C_1$$

$$\Rightarrow y = 1.8\cos(2x) + \sin(2x) + 3x\cos(2x)$$

5.)

Kreyszig 3.2/1

$$y''' + 25y' = 0 \qquad \text{OR} \qquad u = y' \to u'' + 25u = 0$$

$y = e^{\lambda x}$

$$(\lambda^3 + 25\lambda)e^{\lambda x} = 0$$

$$\lambda^2 + 25\lambda = 0 \to \lambda_{1,2} = \pm 5$$

$$u(x) = \tilde{A}\cdot e^{5x} + \tilde{B}e^{-5x}$$

$\int dx$

$$y(x) = \tilde{A}/5\, e^{5x} - \tilde{B}/5\, e^{-5x} + C$$

$$\lambda_1 = 0, \quad \lambda_2 = 5, \quad \lambda_3 = -5$$

$$y(x) = A\cdot e^{\overset{\lambda_2}{5x}} + Be^{\overset{\lambda_3}{-5x}} + C\cdot \underbrace{e^{\overset{\lambda_1}{0x}}}_{=1}$$

Kreyszig 3.2/6

$$(D^5 + 8D^3 + 16D)y = 0 \qquad D^n y = \frac{d^n y}{dx^n}$$

$$\lambda^5 + 8\lambda^3 + 16\lambda = 0$$

$\lambda_1 = 0$

$$\lambda \neq 0: \quad \lambda^4 + 8\lambda^2 + 16 = (\lambda^2 + 4)^2 = 0$$

$$\lambda^2 = -4 \quad \begin{cases} \lambda_{2,3} = 2i \\ \lambda_{4,5} = -2i \end{cases}$$

Degenerate!

$$y(x) = A + B_1\cdot e^{2ix} + B_2 x e^{2ix} + C_1 e^{-2ix} + C_2 x e^{-2ix}$$

$$= A + B_1(\cos 2x + i\sin 2x) + B_2 x(\cos 2x + i\sin 2x)$$

$$+ C_1(\cos 2x - i\sin 2x) + C_2 x(\cos 2x - i\sin 2x)$$

$$= A + (B_1 + C_1)\cos 2x + (B_2 + C_2)x\cos 2x$$

$$+ i(B_1 - C_1)\sin 2x + i(B_2 - C_2)x\sin 2x$$

Kreyszig 3.3/1

$$y''' + 3y'' + 3y' + y = e^{x} - x - 1$$

- Homogeneous:

$$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$$

$$(\lambda + 1)^3 = 0 \quad \Rightarrow \quad \lambda_{1,2,3} = -1$$

$$y_h(x) = Ae^{-x} + Bxe^{-x} + Cx^2e^{-x}$$

- Find a particular solution

$$\rightarrow y_p = Ae^{x} + Bx + C$$

$$y_p' = Ae^{x} + B$$

$$y_p'' = Ae^{x} = y_p'''$$

Equation:

$$Ae^{x} + 3Ae^{x} + 3(Ae^{x} + B) + (Ae^{x} + Bx + C) = e^{x} - x - 1$$

$$8Ae^{x} + Bx + 3B + C = e^{x} - x - 1$$

$$\left.\begin{aligned} 8A &= 1 \\ B &= -1 \\ 3B + C &= -1 \end{aligned}\right\} \quad \begin{aligned} A &= 1/8 \\ B &= -1 \\ C &= 2 \end{aligned}$$

So,

$$y_p(x) = 1/8\, e^{x} - x + 2$$

$$\left[\ y(x) = Ae^{-x} + Bxe^{-x} + Cx^2e^{-x} + 1/8\, e^{x} - x + 2\ \right]$$